Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Seung-Kyum Choi	**Email**	schoi@me.gatech.edu
Institute Name	Georgia Institute of Technology			
Address	813 Ferst Drive, Atlanta, GA, 30068			

2. Technology Overview

Title of Proposed Technology	AI-Driven Predictive Maintenance for Intelligent Robotic Systems in Manufacturing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and materials			
	Medium Category	Small Category	Medium Category	Small Category
	Robotics, Automation machinery	Sensor Technologies		
Concept of Proposed Technology	The proposed technology aims to develop an AI-enabled predictive maintenance framework for manufacturing robotic systems that proactively detects, predicts, and prevents potential failures before they lead to system downtime or operational disruptions. The framework integrates multi-sensor data acquisition, imbalanced feature learning, and meta-learning with few-shot adaptation to address practical challenges in real-world robot maintenance, where abnormal fault data are scarce and operational conditions vary significantly across deployments. By leveraging temperature, current, and vibration signals collected from manufacturing robotic systems, the proposed system continuously monitors robot health conditions and provides real-time			

(max 10 pages)

	insights into component degradation, fault progression, and abnormal behaviors. This enables reliability managers to transition from reactive maintenance to a predictive and condition-based maintenance paradigm, significantly improving fleet availability and lifespan. Key Features • Multi-sensor-based health monitoring using temperature, current, and vibration data • Data-level and feature-level fusion for imbalanced fault detection • Meta-learning-based predictive maintenance with few-shot learning capability • Real-time fault prediction for motors, batteries, and motion alignment anomalies • Seamless integration with operational workflows of manufacturing robots
Rationale for Joint R&D Project	The proposed predictive maintenance technology requires close collaboration between an academic research institute and industrial partners to ensure successful deployment in realistic manufacturing robot's operating environments. • The research institute contributes advanced AI, meta-learning, and predictive analytics expertise. • The Korean SME partner provides real robot platforms, sensor integration, and operational data. • Joint development enables validation under diverse industrial conditions and accelerates technology transfer. Expected Benefits of Joint R&D • Reduced robot downtime through early fault prediction • Improved reliability and lifecycle management of robotic fleets • Faster commercialization of AI-enabled maintenance solutions • Strengthened the competitiveness of Korean robotics SMEs
Key Words	1. AI-Enabled Predictive Maintenance 2. Meta-Learning and Few-Shot Learning 3. Manufacturing Robot 4. Multi-Sensor Data Fusion 5. Robotic Reliability Management

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives	The objective of this project is to develop an intelligent predictive maintenance framework that enables manufacturing robots to autonomously assess their health status and predict potential failures under diverse and data-limited operational conditions. Specific objectives include: • Developing multi-sensor data acquisition and preprocessing pipelines for robots' health monitoring • Developing imbalanced feature learning methods for rare fault detection • Developing a meta-learning-based predictive maintenance model with few-shot adaptation • Validating the framework under laboratory and real-world operational scenarios of robots			
R&D Target Specifications	• Fault Prediction Accuracy: ≥ 90% for key fault modes • Early Detection Capability: Predict failures prior to critical breakdown events • Adaptability: Robust performance with limited fault samples (few-shot setting) • System Compatibility: Seamless integration with existing robot platforms			
R&D Plan	**Year 1: Data Acquisition and Preprocessing** • Integrate temperature, current, and vibration sensors into manufacturing robots' platforms • Collect multi-modal datasets under diverse operating conditions • Conduct a literature review on predictive maintenance and imbalanced learning • Apply advanced preprocessing techniques, including stratified sampling and statistical moment matching **Year 2: Intelligent Predictive Maintenance Framework Development** • Develop data-level and feature-level fusion methods for multi-sensor data • Implement Multi-Signals-to-RGB-Image conversion for CNN-based learning • Design and train deep learning models for fault prediction • Develop meta-learning algorithms with few-shot learning capability **Year 3: Demonstration, Optimization, and Validation** • Conduct extensive testing using synthetic and real-world scenarios • Optimize model performance through iterative feedback loops • Validate robustness, accuracy, and adaptability of the framework • Prepare technical documentation and technology transfer materials			

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will lead the development of the core AI algorithms and predictive maintenance framework. Responsibilities include: • Development of imbalanced feature learning and meta-learning algorithms • Design of predictive maintenance system architecture • Simulation-based validation and performance analysis • Technical guidance and joint evaluation with industrial partners
Expected R&R of Korean SME Partner	The Korean SME partner will focus on hardware integration and system-level validation. Responsibilities include: • Integration of sensors and data acquisition systems on manufacturing robots • System integration of AI models with robotic platforms • Execution of laboratory and real-world testing • Support for commercialization and industrial deployment

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed project will deliver an AI-enabled predictive maintenance prototype capable of accurately predicting failures and improving the operational reliability of manufacturing robot systems. Expected impacts include: • Reduced maintenance costs and unplanned downtime • Extended lifespan of robotic assets • Enhanced competitiveness of Korean robotics SMEs • Advancement of AI-driven maintenance technologies for smart manufacturing and logistics

6. References

(1) Xie, T., Huang, X., and Choi, S.-K., “Intelligent Mechanical Fault Diagnosis Using Multi-Sensor Fusion and Convolutional Neural Network,” IEEE Transactions on Industrial Informatics, Manuscript no. TII-21-2231, 2021.

(2) Xie, T., Huang, X., and Choi, S.-K.,“Metric-Based Meta-Learning for Cross-Domain Few-Shot Identification of Welding Defect,” Journal of Computing and Information Science in Engineering (ASME), Vol. 23, No. 3, 030902, 2023.

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization